NL DATA PROFILING Q&A DOCUMENTATION

TEAM NAME: NULL AWARE

TEAM MEMBERS:

SHRINITHI N

SIVA SURYA P

VIJAYA SRI KESAVAN

YALINI B

ABSTRACT

NulAware AI is an AI-powered Natural Language Data Profiling Assistant that enables users to analyze datasets through conversational interactions. The system automatically generates comprehensive data profiling reports from uploaded CSV files and allows users to query dataset characteristics using natural language.

The application combines traditional data profiling techniques with Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and agentic workflows to provide intelligent insights about data quality, missing values, correlations, outliers, duplicates, and statistical summaries.

The system utilizes Gemini 2.5 Flash for natural language understanding and response generation, LangGraph for agent orchestration, ChromaDB for semantic retrieval, and ydata-profiling for automated dataset analysis. Users can also generate interactive visualizations and export professional PDF reports containing AI-generated recommendations and quality assessments.

NulAware AI simplifies exploratory data analysis by allowing both technical and non-technical users to interact with datasets using natural language instead of writing complex analytical code.

PROBLEM STATEMENT

Data analysts and business users often spend significant time performing exploratory data analysis and data quality assessment before building machine learning models or business reports.

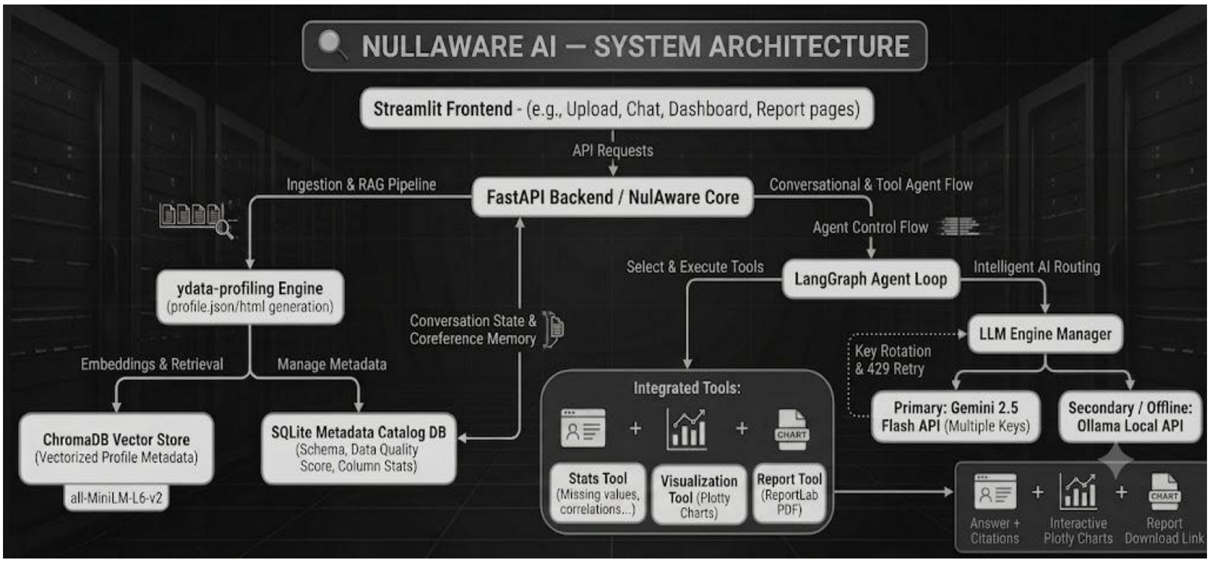
Traditional approaches require knowledge of programming languages, SQL, statistical concepts, and visualization libraries. Non-technical stakeholders face challenges understanding dataset characteristics such as:

- Missing values
- Duplicate records
- Outliers
- Correlations
- Data quality issues
- Distribution patterns

There is a need for an intelligent system that can automatically profile datasets and answer analytical questions using natural language.

NulAware AI addresses this challenge by combining automated profiling, semantic retrieval, agent-based reasoning, and large language models to create an interactive data analysis assistant.

SYSTEM ARCHITECTURE



MODULE DESCRIPTION

1. User Interface Module (Streamlit)

Purpose

The User Interface Module serves as the interaction layer between the user and the system. It provides an intuitive web-based interface for uploading datasets, asking questions, viewing visualizations, and generating reports.

Components

- Upload Page
- Chat Interface
- Dashboard
- Report Generation Page

Functionalities

- Upload CSV datasets
- Display profiling results
- Accept natural language queries

- Show generated charts
- Download PDF reports

Input

- CSV files
- User queries

Output

- Analytical insights
- Visualizations
- Reports

Technology Used

- Streamlit

2. Data Profiling Module

Purpose

This module automatically analyzes uploaded datasets and generates detailed profiling information about data quality, structure, and statistics.

Workflow

CSV Dataset
↓
ydata-profiling
↓
profile.json
profile.html

Functionalities

Dataset Summary

- Number of rows
- Number of columns
- Data types

Statistical Analysis

- Mean
- Median

- Standard deviation
- Variance

Data Quality Analysis

- Missing values
- Duplicate records
- Outliers
- Correlations

Input

- CSV dataset

Output

- profile.json
- profile.html

Technology Used

- ydata-profiling

3. Metadata Extraction Module

Purpose

The profiling report generated by ydata-profiling contains large amounts of information. This module extracts important metadata and converts it into a structured format suitable for semantic search and retrieval.

Functionalities

Extracts:

- Column names
- Data types
- Missing value percentages
- Statistical summaries
- Correlation information
- Outlier information

Workflow

profile.json
↓
extract_profile.py
↓
Structured Metadata

Input

- profile.json

Output

- Structured metadata dictionary

Technology Used

- Python
- JSON Processing

4. Semantic Chunking Module

Purpose

Large metadata documents cannot be directly stored efficiently in vector databases. Therefore, metadata is divided into smaller meaningful chunks.

Functionalities

- Split metadata into logical sections
- Preserve context
- Create retrieval-friendly chunks

Input

- Structured metadata

Output

- Text chunks

Technology Used

- Custom Python Chunking Logic

5. Embedding Generation Module

Purpose

Convert metadata chunks into numerical vector representations that capture semantic meaning.

Workflow

Metadata Chunks
↓
SentenceTransformer
↓
Embeddings

Functionalities

- Text vectorization
- Semantic representation
- Similarity search support

Model Used

all-MiniLM-L6-v2

Input

- Text chunks

Output

- Dense vector embeddings

Technology Used

- Sentence Transformers

6. Vector Database Module (ChromaDB)

Purpose

Stores embeddings and enables semantic retrieval of relevant dataset information.

Functionalities

- Store vectors
- Similarity search
- Context retrieval
- Persistent storage

Workflow

Embeddings
↓
ChromaDB
↓
Top-K Relevant Chunks

Input

- Embeddings

Output

- Relevant metadata chunks

Technology Used

- ChromaDB

7. Retrieval Module (RAG)

Purpose

Implements Retrieval-Augmented Generation (RAG) to provide context-aware answers.

Workflow

User Query
↓
Retriever
↓
Relevant Chunks
↓
Gemini
↓
Answer

Functionalities

- Semantic search
- Context retrieval
- Citation generation
- Improved answer accuracy

Input

- User question

Output

- Retrieved context

Technology Used

- ChromaDB Retriever

8. LLM Management Module

Purpose

Handles communication with Gemini API and manages multiple API keys.

Functionalities

Query Processing

- Sends prompts to Gemini

Multi-Key Management

- API key rotation
- Quota handling
- Retry mechanism

Response Generation

- Natural language answers
- Insight generation

Workflow

Question
↓
Gemini Manager
↓
Gemini API
↓
Response

Technology Used

- Gemini 2.5 Flash
- Google Generative AI SDK

9. Agent Orchestration Module (LangGraph)

Purpose

Acts as the central brain of the system by coordinating retrieval, tools, memory, and response generation.

Components

Router

Selects the appropriate tool.

Memory Manager

Maintains conversation history and resolves follow-up questions.

Functionalities

- Intent classification
- Tool routing
- Context management
- Multi-step reasoning

Technology Used

- LangGraph

10. Visualization Module

Purpose

Generates interactive visual representations of dataset characteristics.

Functionalities

Histogram

Distribution analysis

Scatter Plot

Relationship analysis

Box Plot

Outlier visualization

Bar Chart

Category comparison

Heatmap

Correlation analysis

Workflow

User Request
↓
Visualization Tool
↓
Plotly Figure
↓
Streamlit Display

Technology Used

- Plotly

11. Report Generation Module

Purpose

Generates professional PDF reports summarizing dataset quality and AI-generated insights.

Contents

1. Cover Page
2. Dataset Summary
3. Missing Values Analysis
4. Correlation Analysis
5. Outlier Detection
6. Duplicate Analysis
7. Charts
8. Data Quality Score
9. AI Recommendations

Workflow

Dataset Metadata
↓
AI Summary
↓
Report Builder
↓
PDF Report

Technology Used

- ReportLab

12. Database Module (SQLite)

Purpose

The Conversation Memory Module maintains the context of the current user interaction and enables follow-up questions during a single application session.

Unlike traditional chatbot systems, NulAware AI does not require user authentication or persistent user accounts. Therefore, conversation history is maintained only for the duration of the active Streamlit session.

Functionality

The module stores:

- User questions
- Assistant responses
- Context from previous interactions
- References to previously discussed columns and statistics

This allows the system to resolve follow-up queries and coreferences.

Working Mechanism

User Query

↓

Store in Session Memory

↓

Retrieve Previous Context

↓

Resolve References

↓

Generate Context-Aware Response

Memory Scope

The conversation history is maintained only while the application is running and the page remains active.

The stored memory is automatically cleared when:

- The browser page is refreshed
- The Streamlit application is restarted
- A new application session is created

Technology Used

- Streamlit Session State
- Python Data Structures

13. Tech Stack Table

Layer	Technology
Frontend	Streamlit
Backend	Python
LLM	Gemini 2.5 Flash
Agent	LangGraph
Embeddings	Sentence Transformers
Vector Database	ChromaDB
Database	SQLite
Profiling	ydata-profiling
Charts	Plotly
Reports	ReportLab

CONCLUSION

NulAware AI demonstrates how modern AI technologies can simplify data exploration and profiling. By integrating automated profiling, semantic retrieval, agentic workflows, and large language models, the system provides an intuitive interface for understanding datasets through natural language interactions.

The platform reduces the complexity of exploratory data analysis, improves accessibility for non-technical users, and accelerates the process of discovering valuable insights from data. With support for intelligent querying, visual analytics, and automated report generation, NulAware AI serves as a comprehensive solution for data quality assessment and dataset understanding.